

PART A — MODULE 1 QUESTIONS

MATH1-M1-001

Prompt:

If $3x + 7 = 25$, what is the value of x ?

- A) 4
- B) 6
- C) 8
- D) 10

MATH1-M1-002

Prompt:

A school store has 80 notebooks in stock. If 35% of the notebooks are blue, how many blue notebooks are in stock?

- A) 24
- B) 28
- C) 32
- D) 35

MATH1-M1-003

Prompt:

The numbers of miles five students walked in one week are 4, 6, 8, 10, and 12. What is the mean number of miles walked?

- A) 6
- B) 7
- C) 8
- D) 10

MATH1-M1-004

Prompt:

A linear function f passes through the points (2, 5) and (6, 13). What is the slope of the graph of $y = f(x)$?

- A) 1
- B) 2
- C) 3
- D) 4

MATH1-M1-005

Prompt:

A gym charges a one-time sign-up fee of \$15 plus \$8 for each class attended. Which equation gives the total cost C , in dollars, for attending n classes?

- A) $C = 8n + 15$
- B) $C = 15n + 8$
- C) $C = 23n$
- D) $C = 15n - 8$

MATH1-M1-006

Prompt:

What is the solution (x, y) to the system of equations?

$$\begin{aligned}x + y &= 11 \\x - y &= 3\end{aligned}$$

- A) (4, 7)
- B) (5, 6)
- C) (7, 4)
- D) (8, 3)

MATH1-M1-007

Prompt:

Which inequality represents all values of x that are greater than or equal to 4?

- A) $x > 4$
- B) $x \geq 4$
- C) $x < 4$
- D) $x \leq 4$

MATH1-M1-008

Prompt:

Which expression is equivalent to $2^3 \times 2^4$?

- A) 2^7
- B) 2^{12}
- C) 4^7
- D) 4^{12}

MATH1-M1-009

Prompt:

Which expression is equivalent to $5(x + 3) - 2x$?

- A) $3x + 3$
- B) $3x + 15$
- C) $7x + 15$
- D) $5x + 1$

MATH1-M1-010

Prompt:

A bag contains 6 red marbles, 5 blue marbles, and 4 green marbles. If one marble is selected at random, what is the probability that the marble selected is blue?

- A) $1/5$
- B) $1/4$
- C) $1/3$
- D) $2/5$

MATH1-M1-011

Prompt:

A rectangle has a length of 9 inches and a width of 4 inches. What is the area, in square inches, of the rectangle?

- A) 13
- B) 26
- C) 36
- D) 72

MATH1-M1-012

Prompt:

A right triangle has legs with lengths 5 and 12. What is the length of the hypotenuse?

- A) 11

- B) 13
- C) 17
- D) 30

MATH1-M1-013

Prompt:

The equation $x^2 - 9 = 0$ has two solutions. What is the positive solution?

- A) 1
- B) 3
- C) 6
- D) 9

MATH1-M1-014

Prompt:

If $f(x) = 2x^2 - 3$, what is $f(4)$?

- A) 13
- B) 19
- C) 29
- D) 35

MATH1-M1-015

Prompt:

The ratio of boys to girls in a club is 3 to 5. If there are 24 boys in the club, how many girls are in the club?

- A) 32
- B) 36
- C) 40
- D) 48

MATH1-M1-016

Prompt:

The table shows the number of cups of flour needed for different numbers of batches of muffins.

Batches: 2, 3, 5, 6

Cups of flour: 4, 6, 10, 12

Based on the table, how many cups of flour are needed for 8 batches?

- A) 14
- B) 16
- C) 18
- D) 20

MATH1-M1-017

Prompt:

The circumference of a circle is 18π inches. What is the radius, in inches, of the circle?

- A) 6
- B) 9
- C) 18
- D) 36

MATH1-M1-018

Prompt:

Which expression is equivalent to $\sqrt{49} + \sqrt{16}$?

- A) 9
- B) 11
- C) 13
- D) 65

MATH1-M1-019

Prompt:

Two angles are supplementary. One angle measures 68° . What is the measure, in degrees, of the other angle?

- A) 22
- B) 68
- C) 112
- D) 122

MATH1-M1-020

Prompt:

The value of a computer is modeled by the equation $V = 900 - 120t$, where V is the value, in dollars, t years after purchase. What does the number 120 represent in this model?

- A) The computer's initial value, in dollars
- B) The number of years after purchase
- C) The decrease in value, in dollars, each year
- D) The computer's value after 120 years

MATH1-M1-021

Prompt:

A rectangular prism has a length of 6 centimeters, a width of 5 centimeters, and a height of 4 centimeters. What is the volume, in cubic centimeters, of the prism?

- A) 60
- B) 90
- C) 120
- D) 150

MATH1-M1-022

Prompt:

The table shows the numbers of students in a music program who play piano or guitar.

Instrument: Piano, Guitar

Freshmen: 12, 18

Sophomores: 16, 14

How many sophomores are in the music program?

- A) 14
- B) 16
- C) 30
- D) 60

PART B — MODULE 2 QUESTIONS

MATH1-M2-001

Prompt:

The system of equations has solution (a, b) .

$$2x + y = 17$$

$$x - y = 4$$

What is the value of $a + b$?

- A) 7
- B) 10
- C) 13
- D) 17

MATH1-M2-002

Prompt:

The function h is defined by $h(x) = x^2 - 6x + 11$. What is the minimum value of $h(x)$?

- A) 1
- B) 2
- C) 3
- D) 11

MATH1-M2-003

Prompt:

Which expression is equivalent to $(3x^2y)(4xy^3)$?

- A) $7x^3y^4$
- B) $12x^2y^3$
- C) $12x^3y^4$
- D) $12x^3y^3$

MATH1-M2-004

Prompt:

A population of bacteria doubles every hour. If the population is 300 at noon, which expression gives the population 5 hours after noon?

- A) $300 + 2(5)$
- B) 300×2^5
- C) 300×5^2
- D) 2×300^5

MATH1-M2-005

Prompt:

A line passes through the points $(-2, 9)$ and $(4, -3)$. What is the y -intercept of the line?

- A) -5
- B) 1
- C) 3
- D) 5

MATH1-M2-006

Prompt:

A jacket originally costs \$80. During a sale, the price is reduced by 25%. A customer also uses a coupon for \$10 off the sale price. What is the final price of the jacket?

- A) \$50
- B) \$55

- C) \$60
- D) \$70

MATH1-M2-007

Prompt:

A rectangle has a perimeter of 34 units. Its length is 5 units greater than its width. What is the area, in square units, of the rectangle?

- A) 66
- B) 72
- C) 84
- D) 90

MATH1-M2-008

Prompt:

If $g(x) = 3x - 2$ and $g(k) = 19$, what is the value of k ?

- A) 5
- B) 6
- C) 7
- D) 8

MATH1-M2-009

Prompt:

The graph of $y = (x - 2)(x + 5)$ crosses the x -axis at two points. What is the greater x -coordinate of these two points?

- A) -5
- B) -2
- C) 2
- D) 5

MATH1-M2-010

Prompt:

Which expression is equivalent to $x^2 + 8x + 15$?

- A) $(x + 3)(x + 5)$
- B) $(x + 1)(x + 15)$
- C) $(x - 3)(x - 5)$
- D) $(x + 4)(x + 4)$

MATH1-M2-011

Prompt:

A recipe uses 2 cups of oats for every 3 cups of flour. If a baker uses 15 cups of flour, how many cups of oats are needed?

- A) 8
- B) 10
- C) 12
- D) 18

MATH1-M2-012

Prompt:

The data set 6, 8, 10, 12, 14 has a median of 10. If the number 22 is added to the data set, what is the median of the new data set?

- A) 10

- B) 11
- C) 12
- D) 14

MATH1-M2-013

Prompt:

A circle in the xy -plane has center $(3, -2)$ and radius 5. Which equation represents this circle?

- A) $(x - 3)^2 + (y + 2)^2 = 25$
- B) $(x + 3)^2 + (y - 2)^2 = 25$
- C) $(x - 3)^2 + (y + 2)^2 = 5$
- D) $(x + 3)^2 + (y - 2)^2 = 5$

MATH1-M2-014

Prompt:

In a right triangle, one acute angle has sine $5/13$. If the length of the hypotenuse is 26, what is the length of the side opposite this angle?

- A) 5
- B) 10
- C) 13
- D) 24

MATH1-M2-015

Prompt:

The table shows values of a linear function f .

x : $-1, 0, 2, 5$
 $f(x)$: $7, 5, 1, -5$

Which equation defines f ?

- A) $f(x) = -2x + 5$
- B) $f(x) = 2x + 5$
- C) $f(x) = -x + 6$
- D) $f(x) = -3x + 4$

MATH1-M2-016

Prompt:

What is the solution to the equation $\sqrt{x + 7} = 5$?

- A) 12
- B) 18
- C) 25
- D) 32

MATH1-M2-017

Prompt:

For what value of c does the equation $4x + c = 4x + 9$ have no solution?

- A) 0
- B) 4
- C) 9
- D) Any value except 9

MATH1-M2-018

Prompt:

A survey asked students whether they prefer morning or afternoon study sessions. The results are shown.

Preference: Morning, Afternoon

Grade 10: 18, 12

Grade 11: 10, 20

If one student from the survey is selected at random from the students who prefer afternoon sessions, what is the probability that the student is in Grade 11?

- A) $\frac{1}{3}$
- B) $\frac{3}{8}$
- C) $\frac{5}{8}$
- D) $\frac{2}{3}$

MATH1-M2-019

Prompt:

A cube has side length 3 inches. A larger cube has side length 6 inches. The volume of the larger cube is how many times the volume of the smaller cube?

- A) 2
- B) 4
- C) 6
- D) 8

MATH1-M2-020

Prompt:

If $3^x = 81$, what is the value of $x + 2$?

- A) 4
- B) 5
- C) 6
- D) 8

MATH1-M2-021

Prompt:

The function p is defined by $p(x) = 2x^2 + bx + 8$, where b is a constant. If $p(2) = 30$, what is the value of b ?

- A) 3
- B) 5
- C) 7
- D) 9

MATH1-M2-022

Prompt:

A theater sold adult tickets for \$12 each and student tickets for \$8 each. A total of 75 tickets were sold, and the total revenue was \$780. How many student tickets were sold?

- A) 25
- B) 30
- C) 35
- D) 45

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH1-M1-001 — Correct: 6 — Explanation: Subtract 7 from both sides to get $3x = 18$, then divide by 3 to get $x = 6$.

MATH1-M1-002 — Correct: 28 — Explanation: 35% of 80 is $0.35 \times 80 = 28$.

MATH1-M1-003 — Correct: 8 — Explanation: The sum is $4 + 6 + 8 + 10 + 12 = 40$, and $40 \div 5 = 8$.

MATH1-M1-004 — Correct: 2 — Explanation: The slope is $(13 - 5)/(6 - 2) = 8/4 = 2$.

MATH1-M1-005 — Correct: $C = 8n + 15$ — Explanation: The total cost is the fixed sign-up fee, 15, plus 8 dollars for each of n classes.

MATH1-M1-006 — Correct: (7, 4) — Explanation: Adding the equations gives $2x = 14$, so $x = 7$. Then $7 + y = 11$, so $y = 4$.

MATH1-M1-007 — Correct: $x \geq 4$ — Explanation: The phrase “greater than or equal to” is represented by the symbol $\geq$.

MATH1-M1-008 — Correct: 2^7 — Explanation: When multiplying powers with the same base, add exponents: $2^3 \times 2^4 = 2^7$.

MATH1-M1-009 — Correct: $3x + 15$ — Explanation: Distribute 5 to get $5x + 15 - 2x$, which simplifies to $3x + 15$.

MATH1-M1-010 — Correct: $1/3$ — Explanation: There are 15 marbles total and 5 are blue, so the probability is $5/15 = 1/3$.

MATH1-M1-011 — Correct: 36 — Explanation: The area of a rectangle is length $\times$ width, so $9 \times 4 = 36$.

MATH1-M1-012 — Correct: 13 — Explanation: By the Pythagorean theorem, the hypotenuse is $\sqrt{(5^2 + 12^2)} = \sqrt{169} = 13$.

MATH1-M1-013 — Correct: 3 — Explanation: $x^2 - 9 = 0$ gives $x^2 = 9$, so $x = 3$ or $x = -3$. The positive solution is 3.

MATH1-M1-014 — Correct: 29 — Explanation: Substitute 4 for x : $f(4) = 2(4^2) - 3 = 32 - 3 = 29$.

MATH1-M1-015 — Correct: 40 — Explanation: The ratio boys:girls is 3:5. Since 24 boys represents 3 parts, each part is 8, so girls = $5 \times 8 = 40$.

MATH1-M1-016 — Correct: 16 — Explanation: The table shows 2 cups of flour per batch, so 8 batches require $8 \times 2 = 16$ cups.

MATH1-M1-017 — Correct: 9 — Explanation: Circumference is $2\pi r$. Since $2\pi r = 18\pi$, $r = 9$.

MATH1-M1-018 — Correct: 11 — Explanation: $\sqrt{49} = 7$ and $\sqrt{16} = 4$, so the sum is 11.

MATH1-M1-019 — Correct: 112° — Explanation: Supplementary angles have a sum of 180° , so the other angle is $180 - 68 = 112^\circ$.

MATH1-M1-020 — Correct: The decrease in value, in dollars, each year — Explanation: The coefficient of t is -120 , so the value decreases by 120 dollars for each increase of 1 year.

MATH1-M1-021 — Correct: 120 — Explanation: The volume is length $\times$ width $\times$ height = $6 \times 5 \times 4 = 120$.

MATH1-M1-022 — Correct: 30 — Explanation: The number of sophomores is $16 + 14 = 30$.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH1-M2-001 — Correct: 10 — Explanation: Add the equations to get $3x = 21$, so $x = 7$. Then $7 - y = 4$, so $y = 3$. Thus $a + b = 7 + 3 = 10$.

MATH1-M2-002 — Correct: 2 — Explanation: Complete the square: $x^2 - 6x + 11 = (x - 3)^2 + 2$. Since $(x - 3)^2 \geq 0$, the minimum value is 2.

MATH1-M2-003 — Correct: $12x^3y^4$ — Explanation: Multiply coefficients and add exponents of like bases: $(3 \times 4)x^{2+1}y^{1+3} = 12x^3y^4$.

MATH1-M2-004 — Correct: 300×2^5 — Explanation: Doubling each hour means multiplying by 2 for each hour. After 5 hours, the population is 300×2^5 .

MATH1-M2-005 — Correct: 5 — Explanation: The slope is $(-3 - 9)/(4 - (-2)) = -12/6 = -2$. Using $y = -2x + b$ and point $(4, -3)$, $-3 = -8 + b$, so $b = 5$.

MATH1-M2-006 — Correct: \$50 — Explanation: A 25% reduction from \$80 is \$20, so the sale price is \$60. Applying the \$10 coupon gives \$50.

MATH1-M2-007 — Correct: 66 — Explanation: Let the width be w and the length be $w + 5$. Then $2w + 2(w + 5) = 34$, so $4w + 10 = 34$ and $w = 6$. The length is 11, so the area is 66.

MATH1-M2-008 — Correct: 7 — Explanation: Since $g(k) = 19$, $3k - 2 = 19$. Then $3k = 21$, so $k = 7$.

MATH1-M2-009 — Correct: 2 — Explanation: The x -intercepts occur when $(x - 2)(x + 5) = 0$, so $x = 2$ or $x = -5$. The greater value is 2.

MATH1-M2-010 — Correct: $(x + 3)(x + 5)$ — Explanation: The factors must multiply to 15 and add to 8, so $x^2 + 8x + 15 = (x + 3)(x + 5)$.

MATH1-M2-011 — Correct: 10 — Explanation: The ratio of oats to flour is 2:3. For 15 cups of flour, the scale factor is 5, so oats = $2 \times 5 = 10$ cups.

MATH1-M2-012 — Correct: 11 — Explanation: The new ordered set is 6, 8, 10, 12, 14, 22. The median is the average of the third and fourth values: $(10 + 12)/2 = 11$.

MATH1-M2-013 — Correct: $(x - 3)^2 + (y + 2)^2 = 25$ — Explanation: A circle with center (h, k) and radius r has equation $(x - h)^2 + (y - k)^2 = r^2$. Here $h = 3$, $k = -2$, and $r^2 = 25$.

MATH1-M2-014 — Correct: 10 — Explanation: Sine is opposite/hypotenuse. Since $5/13 = \text{opposite}/26$, the opposite side length is $26 \times 5/13 = 10$.

MATH1-M2-015 — Correct: $f(x) = -2x + 5$ — Explanation: From $x = -1$ to $x = 0$, $f(x)$ decreases by 2, so the slope is -2 . Since $f(0) = 5$, the equation is $f(x) = -2x + 5$.

MATH1-M2-016 — Correct: 18 — Explanation: Square both sides: $x + 7 = 25$, so $x = 18$.

MATH1-M2-017 — Correct: Any value except 9 — Explanation: Subtracting $4x$ from both sides gives $c = 9$. If c is any value other than 9, the equation is false and has no solution.

MATH1-M2-018 — Correct: $5/8$ — Explanation: There are $12 + 20 = 32$ students who prefer afternoon sessions. Of these, 20 are in Grade 11, so the probability is $20/32 = 5/8$.

MATH1-M2-019 — Correct: 8 — Explanation: Volume scales by the cube of the side-length scale factor. Since $6/3 = 2$, the volume scale factor is $2^3 = 8$.

MATH1-M2-020 — Correct: 6 — Explanation: Since $81 = 3^4$, $x = 4$. Therefore, $x + 2 = 6$.

MATH1-M2-021 — Correct: 7 — Explanation: Substitute $x = 2$ into $p(x)$: $2(2^2) + 2b + 8 = 30$. This gives $8 + 2b + 8 = 30$, so $2b = 14$ and $b = 7$.

MATH1-M2-022 — Correct: 30 — Explanation: Let s be the number of student tickets. Then adult tickets are $75 - s$. The revenue equation is $8s + 12(75 - s) = 780$. This simplifies to $900 - 4s = 780$, so $s = 30$.